Matias Guillen

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Education

University of Central Florida — Orlando, FL
Bachelor of Science in Electrical Engineering | GPA: 3.4 — Expected May 2027
Relevant Coursework: Electronics 1 & 2, Electromagnetic Fields, Semiconductor Devices, Real Time Systems

Experience

Electrical Engineering Antenna and RF Intern — May 2026 – August 2026
Lockheed Martin RMS — Moorestown, NJ

- Programmed in C on a Linux-based real-time RF safety system to acquire and transmit serial I/O data, using a logic analyzer, oscilloscope and synthesized signal generator to validate a bit-exact match to the legacy FPGA being replaced.
- Configured 4 peripheral modules via TI SysConfig, cross-referencing device datasheets to emulate FPGA behavior on a TI board and reproduce peak and average RF power detector outputs in agreement with calibration data of the system.
- Developed test plans and review documentation for near-field calibration of S-band phased array antennas, tracing requirements in DOORS to ensure test coverage aligned with system specifications across software and hardware.

Production Test Engineering CWEP — April 2024 – April 2026
Lockheed Martin MFC - College Work Experience Program (CWEP) — Orlando, FL

- Conducted component-level failure analysis on 20 sensor hardware units monthly using multimeters and test equipment, and engineered 12 custom test fixtures using SolidWorks to mitigate moisture interference, improving RF signal integrity.
- Managed the review process for 160 certified test documents in Windchill, ensuring clear and concise work instructions for 34 Electronics Test Specialists across 8 program areas, supporting 35 unique hardware builds and 15 USG and allied customers.
- Ensured 100% operational readiness for 4,500 annual test operations by managing the calibration schedule for 135 test assets in Indysoft, coordinating with the metrology lab to prioritize single-point failures across 9 contract requirements.

Projects

Senior Design - Crew Environmental Monitoring System - Electrical Designer — April 2026 – Present

- Developing requirements for a NASA crew environmental monitoring system using sensor fusion across O2, CO, temperature, and humidity sensors to enable trend analysis and early hazard prediction, improving crew safety.

IEEE UCF - Internal Project Competition - Project Lead — Apr 2024 – Apr 2025

- Led 20 members across 9 teams in a multi-level (beginner to advanced) project competition, providing technical mentorship through the full engineering design process from concept to prototype and culminating in a final showcase of their new skills.
- Developed 9 workshops on technical topics including programming in Arduino, KiCad, and mechanical design in Onshape, which equipped members with essential skills to successfully execute their component research and design.
- Managed a $300 project budget across 9 teams, optimizing resource allocation by advising on cost-effective design changes that also improved overall system efficiency and tracking components in a BOM.

IEEE SoutheastCon Hardware Competition - Electrical and Mechanical - Member — Sep 2023 – Apr 2024

- Owned the end-to-end design of a modular power system in KiCad, featuring multiple voltage rails (5V and 12V) and overcurrent protection, which accelerated prototyping by 5 weeks by enabling parallel hardware and software development.
- Developed validation of power components and mechanical subsystems (zip-line mechanism and pinch arm) in Fusion 360, leading to the identification and resolution of 3 critical design flaws and ensuring successful final system integration.

Leadership

IEEE Eta Kappa Nu (HKN) – Zeta Chi – Honor Society
Chapter Vice President & Chapter Officer — Apr 2025 – Present

- Led the relaunch of the UCF HKN chapter by organizing and speaking at 3 induction ceremonies across 2025–2026, establishing 46 verified members while upholding the academic standards and integrity central to HKN's mission.

IEEE UCF
President/Student Branch Chair — Apr 2025 – Apr 2026

- Oversaw the largest electrical and computer engineering club at UCF, directing a board of 13 officers across 6 functional areas (including workshops, projects, and outreach) growing membership 64% from 277 to 455 active members over a year.
- Drove club direction by securing $10,000+ in sponsorships from industry partners including Northrop Grumman and Burns & McDonnell, facilitating 30+ events that connected members with career opportunities in engineering disciplines.
- Led the Circuit Design Competition team to 2nd place among 39 competing colleges at IEEE SoutheastCon 2026 while driving the organizational initiatives that earned the branch the IEEE Exemplary Student Branch Award.

Project Chair — Apr 2024 – Apr 2025

- Coordinated with 8 project leads in charge of 6 different technical areas by providing design assistance, weekly budget meetings and project organization to facilitate a strong impactful image and engagement with projects.
- Developed and implemented a formal budget approval process requiring project leads to submit detailed proposals with a Bill of Materials (BOM), enhancing spending transparency and successfully allocating $1900 to 12 project budget requests.

Skills

EDA & Design: KiCad, LTSpice, Fusion 360, SolidWorks (CSWA), Windchill
Hardware & Test: Oscilloscope, Multimeter, Logic Analyzer, Synthesized Signal Generator, PCB Fab & Assembly, Soldering
Programming & Industry Tools: C, C++, MATLAB & Simulink, Python, Linux, Git, Verilog